WASP-2b

R-0194

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_180819` · created 2026-07-16T21:18:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458349.7744 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.027
Transit depth (Rp/Rs)^2	2.57 +/- 0.87 [%]
Orbital Inclination (inc)	89.56 +/- 1.73
Airmass coefficient 1 (a1)	1.332 +/- 0.04
Airmass coefficient 2 (a2)	-0.24 +/- 0.11
Scatter in the residuals of the lightcurve fit is	3.06 %
Best Comparison Star	#3 - [418, 336]
Optimal Aperture	3.07
Optimal Annulus	12.63
Transit Duration (day)	0.0978 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.027 vs 0.1326 (deviation 0.78σ)
Duration fit vs published	0.0978 d vs 0.0737 d (deviation 4.28σ)
Mid-time O–C	13.7 min
Depth SNR	4.5
Residual scatter	3.06 %

Light curve

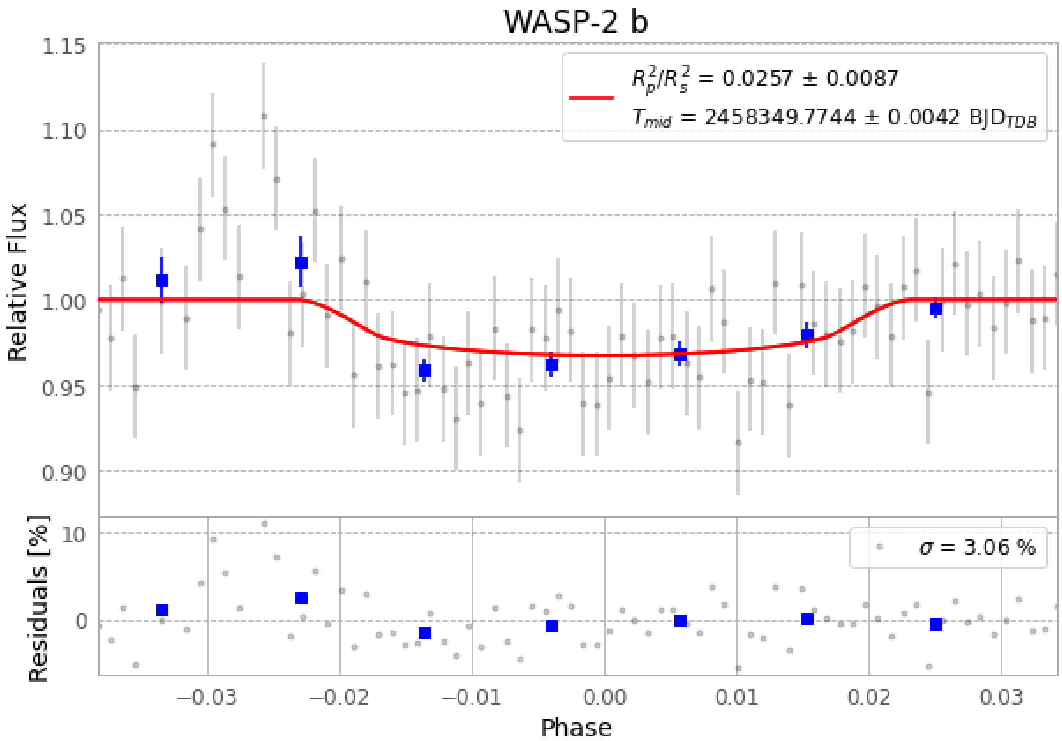

Figure 1 — FinalLightCurve_WASP-2 b_2018-08-18.png (SHA-256 873e5ebee398ab83…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 220], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6b31556afe3e6927…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d`

Data provenance

76 FITS frames (50 MB), `WASP-2180819042712.FITS` → `WASP-2180819081212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180819.log (9f199aae1775…), FinalLightCurve_WASP-2 b_2018-08-18.png (873e5ebee398…), FinalLightCurve_WASP-2 b_2018-08-18.csv (a35a680ad5f2…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

